A12749W1

SECOND PUBLIC EXAMINATION

Honour School of Philosophy, Politics and Economics
Honour School of Economics and Management
Honour School of History and Economics
Honour School of Engineering, Economics and Management (Part II)
Honour School of Materials, Economics and Management (Part I)

MICROECONOMICS

TRINITY TERM 2016

Friday 27th May 2016, 09:30 - 12:30

Please start the answer to each question on a separate page.

There are 10 questions in this paper.

Answer **all** *questions in Part A and* **two** *long questions from Part B.*
Part A attracts 1/3 of the marks in total; each question in Part B attracts 1/3 of the marks.

For the questions in Part A, and the multipart question in Part B, the weight assigned to each part is indicated in square brackets.

Candidates may use their own calculators.

Do not turn over until told that you may do so.

Part A

Numbers in square brackets indicate marks as a percentage of the total for Part A

1. "When firms compete in prices, prices fall to marginal cost and the firms make no profits."

   (a) Describe carefully a game-theoretic model of a price-setting duopoly that illustrates this statement. [10%]

   (b) Briefly describe a modification of your model in which the statement would not hold. Explain intuitively why in this case firms are able to make profits in equilibrium. [10%]

2. (a) Define the risk premium associated with a lottery. [5%]

   Janet has a ticket for a lottery that will leave her with final wealth of either 10 or 100, each equally likely; there is another lottery that (if she owned a ticket) would leave Janet with final wealth of either 35 or 65, again each equally likely, but it is Sam that has a ticket for this second lottery.

   (b) If Janet is willing to exchange tickets with Sam, what does this say about Janet's attitude to risk? [9%]

   (c) Sam does not agree to a straight exchange of tickets, but requires Janet to pay a price p in addition. If Janet derives utility $\ln w$ from final wealth w, what is the highest value of p that she is prepared to pay? [9%]

3. Three bicycles are available: one each of high, medium and low quality. There are three type X consumers, two type Y consumers and two type Z consumers where each consumer-type values each quality of bicycle differently:

Quality	Type X value	Type Y value	Type Z value
High	90	100	115
Medium	35	60	50
Low	25	20	30

   Everyone knows the value that each type of consumer attaches to each quality of bicycle.

   No-one wants more than one bicycle, and the price at which any bicycle is traded will be determined by supply and demand.

   (a) Suppose that type X consumers initially own the bicycles – one each. If *all* the consumers know the quality of each bicycle, which consumer-type will end up owning which bicycle?

   Is the equilibrium price for each bicycle fully determined? [8%]

From now on, assume that buyers do not know the quality of any particular bicycle for sale, but sellers do know the quality of the one that they own.

(b) Continue to suppose that type X consumers initially have one bicycle each. Is there adverse selection in this market? What will be the equilibrium outcome? [10%]

(c) The market in part (b) has closed, and any new owner of a bicycle now knows the quality they bought. Suppose that a type Z consumer bought the low quality bicycle in that market. If the market were to reopen for a second round of trading, describe what the equilibrium outcome would be. [8%]

4. A company has purchased the rights to a very large number of music tracks, and incurs no further costs when tracks are downloaded and played. There is a large number, N, of consumers, who can play a track at price p. Consumers have quasi-linear utility: their net benefit from playing x tracks is

$$B(x) = ax - bx^2 - h - px$$

where a and b are parameters such that $a^2 < 4b$. Subscription is free, but h is the initial hassle (disutility) of subscribing to the service. Consumers differ in the level of hassle h that they face: h is uniformly distributed between 0 and 1.

(a) Find the number of tracks, x^*, that a subscriber of type h will choose to play at price $p < a$, and the net benefit that the subscriber will obtain. [8%]

(b) Show that the number of consumers who will subscribe is $N^* = N\dfrac{(a-p)^2}{4b}$. [4%]

(c) Show that the profit-maximising price for the firm is $p^* = a/4$. (There is no need to check the second order condition.) How many consumers will subscribe? [8%]

(d) What would be the efficient price level, and how many consumers would subscribe at that price? [5%]

(e) Discuss, without further calculations, the comparison between welfare (consumer and producer surplus) at the profit-maximising price level, and welfare at the efficient price level. [6%]

Part B

5. "Increasing returns to scale are pervasive in the modern economy. Some are internal to firms, others take the form of spillovers (positive reciprocal externalities) between firms."

   What problems does this pose for the theory of general competitive equilibrium? How should our understanding of the welfare effects of policy be adapted to incorporate increasing returns to scale?

6. A group of countries is seeking to take collective action to reduce greenhouse gas emissions. What policy scheme would you advise them to adopt? What difference would it make if many countries were outside the group?

7. When considering a proposed merger between two or more firms, how would a competition authority determine the size of the market within which the merging firms operate? Having determined the size of the market, how would the competition authority assess whether the merger should be allowed?

8. "If everyone in an economy had the same level of wealth and the same attitude to risk, then there would be no scope for insurance."

   Do you agree?

9. A risk-neutral principal wants to hire an agent to undertake a project. What considerations does the principal take into account when designing the optimal contract? (For illustration purposes, you may assume that there are just two possible levels of gross profit, and two levels of effort that the agent can exert.)

   Compare the optimal risk-sharing contracts when effort is observable and when it is not: if the principal wants to induce different effort levels in these two situations, who is better or worse off? If on the other hand the principal wants to induce the same effort level in these two situations, is anyone better or worse off?

10. Suppose that two firms engage in Cournot competition. The firms simultaneously choose individual production $q_i \geq 0$ $(i = 1, 2)$. The market clears at a price $p = 100 - Q$, where $Q = q_1 + q_2$. Each firm has a cost function $C_i(q_i) = 10q_i$.

(a) Find the unique pure-strategy Cournot-Nash equilibrium of this game. How much profit does each firm earn in the equilibrium? [15%]

Now suppose that before the two firms engage in Cournot competition, firm 1 can invest in cost reduction, but firm 2 cannot. If firm 1 invests, it incurs a cost of $K > 0$ and its cost function in the Cournot competition drops to $C_i(q_i) = 4q_i$. Firm 2 discovers whether firm 1 invested or not before Cournot competition begins.

(b) If firm 1 chooses to invest, what will be the Cournot-Nash equilibrium of the quantity-setting game? How much profit does each firm earn in the equilibrium (ignoring any prior investment cost of firm 1)? [15%]

(c) Explain intuitively why firm 1 earns more than firm 2 in the Cournot-Nash equilibrium found in part (b). [8%]

(d) Consider a subgame-perfect Nash equilibrium of the two-stage game in which: (i) firm 1 chooses to invest or not in the first stage; and (ii) the firms engage in Cournot competition in the second stage after observing firm 1's investment decision in the first stage. For what values of K will firm 1 invest? Describe fully the strategies in the subgame-perfect Nash equilibrium in which firm 1 chooses to invest. [25%]

(e) Suppose that $K = 200$. Show that there is a Nash equilibrium of the two-stage game in which firm 1 does not invest in the first stage and neither firm conditions its production level on whether firm 1 invests or not in the first stage. [25%]

(f) Using your answers to parts (d) and (e), explain how subgame-perfect Nash equilibrium relates to the notion of credibility of agents' play in games. [12%]